FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 23-09-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. At t = 10 sec

$$x_{cm} = 100 \times \frac{4}{5} \times 10 = 800 \text{ m}$$

$$y_{cm} = 100 \times \frac{3}{5} \times 10 - 5 \times 100 = 100 \text{ m}$$

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$800 = \frac{m \times 400 + 2m x_2}{3m}$$

$\Rightarrow x_2 = 1000$ m

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$100 = \frac{m \times 60 + 2m y_2}{3m}$$

$\Rightarrow y_2 = 120$ m

2. B

Sol. $f_P = \frac{3v_P}{4\ell_P}$, $f_Q = \frac{3v_{mix}}{2\ell_Q}$

$$\frac{f_P}{f_Q}=\frac{\ell_Q}{2\ell_P}\frac{v_{Hz}}{v_{mix}}\sqrt{\frac{\gamma_{H_2}}{\gamma_{mix}}\times\frac{T_{Hz}}{T_{mix}}\times\frac{M_{mix}}{M_{H_2}}}=\sqrt{\frac{7}{5}\times\frac{2}{3}\times\frac{400}{420}\times\frac{3}{2}}=\frac{2}{\sqrt{3}}$$

3. C

Sol. In equilibrium

$Mg = m\omega^2 r$

When the weight is displaced by a small distance x downwards and released.

From conservation of angular momentum

$m\omega r^2 = m\omega'(r-x)^2$

Acceleration of the weight $a = \frac{Mg-T}{M}=\frac{m\omega^2 r - m\omega'^2(r-x)}{M}$

$$= \frac{m}{M}\left[\omega^2 r-\frac{\omega^2 r^4}{(r-x)^4}(r-x)\right]=\frac{m}{M}\omega^2 r\left[1-\frac{r^3}{(r-x)^3}\right]$$

$$= \frac{m\omega^2 r}{M}\left[1-\left(1-\frac{x}{r}\right)^{-3}\right]=\frac{m\omega^2 r}{M}\left(\frac{-3x}{r}\right)$$

from binomial approximation $\left[\left(1-\frac{x}{r}\right)^{-3}\approx 1+\frac{3x}{r}\ \text{since } x<<r\right]$

$$\Rightarrow a = -\frac{3m\omega^2 x}{M}$$

$$\Rightarrow \text{frequency } f = \frac{1}{2\pi}\sqrt{\frac{3m\omega^2}{M}}$$

4. C

Sol. $x = u\cos\theta t$

$y = u\sin\theta\, t - \frac{1}{2}gt^2$

$v_x = u\cos\theta$, $v_y = u\sin\theta - gt$

The stone is at the greatest distance from the point of projection O. when its velocity is perpendicular to the position vector.

The condition for this is

$$\frac{y}{x}=-\frac{v_x}{v_y}\qquad \vec{v}=u\cos\theta\hat{i}+(u\sin\theta-gt)\hat{j},\quad \vec{r}=u\cos\theta t\hat{i}+\left(u\sin\theta t-\frac{1}{2}gt^2\right)\hat{j}$$

$$\Rightarrow t^2-\frac{3u\sin\theta}{g}t+\frac{2u^2}{g^2}=0$$

If this is not happen, discriminate $D < 0$

$$\Rightarrow \left(\frac{3u\sin\theta}{g}\right)^2<4\frac{2u^2}{g^2}$$

$$\Rightarrow \sin\theta<\sqrt{\frac{8}{9}}$$

$$\sin\theta_{max}=\frac{\sqrt{8}}{3}$$

5. A, C

Sol. The upper ball will ultimately receives the largest possible fraction of total energy if after collision between the two balls m_2 comes to rest.

$$\Rightarrow \frac{(m_2 - m_1)\sqrt{2gh} - 2m_1\sqrt{2gh}}{m_1 + m_2} = 0 \quad \Rightarrow m_2 = 3m_1$$

If $m_2 >> m_1$

$v_1 = -u_1 + 2u_2 = 2gh + 2\sqrt{2gh} = 3\sqrt{2gh}$

$$h_{max} = \frac{v_1^2}{2g} = 9h$$

6. B, C, D

Sol. From conservation of angular momentum

$$(2mx^2 + I)\omega' = (2mR^2 + I)\omega$$

$$\Rightarrow \omega' = \frac{(2mR^2 + I)\omega}{2mx^2 + I}$$

Force exerted by insect on the turntable

$$F = m\omega'^2 x = m\frac{(2mR^2 + I)^2\omega^2}{(2mx^2 + I)^2}x$$

For F to maximum $\frac{dF}{dx} = 0$

$$\Rightarrow x = \sqrt{\frac{I}{6m}} = \sqrt{\frac{\frac{2}{3}mR^2}{6m}} = \frac{R}{3}$$

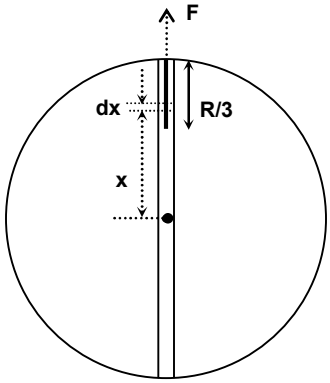
$$\omega' = \left(\frac{2mR^2 + \frac{2}{3}mR^2}{2m\frac{R^2}{9} + \frac{2}{3}mR^2}\right)\omega = 3\omega$$

$$F_{max} = m\omega'^2 x = m \times 9\omega^2 \times \frac{R}{3} = 3m\omega^2 R$$

7. A, B

Sol. (A) Required force F = weight of the chain

$$= \int_{2R/3}^{R} (\lambda dx)\frac{gx}{R} = \frac{5\lambda gR}{18}$$



(B) Initial potential energy of the chain

$$U_i = \int_{2R/3}^{R} -\lambda dx\frac{GM}{2R}\left(3 - \frac{x^2}{R^2}\right)$$

$$= \int -\lambda dx\frac{gR^2}{2R}\left(3 - \frac{x^2}{R^2}\right) = -\frac{31\lambda gR^2}{81}$$

Final potential energy of the chain

$$U_f = \int_{0}^{R/3} -\lambda dx\frac{gR^2}{2R}\left(3 - \frac{x^2}{R^2}\right) = -\frac{40\lambda gR^2}{81}$$

$$\frac{1}{2}mv^2 = U_i - U_f \Rightarrow v = \sqrt{\frac{2gR}{3}}$$

$$U_i = -31\lambda\frac{gR^2}{81}$$

$$U_f = -\frac{GMm}{R} = -\frac{\left(gR^2\lambda\frac{R}{3}\right)}{R} = -\frac{\lambda gR^2}{3}$$

$$W = \Delta U = \frac{4\lambda gR^2}{81}$$

8. A, C

Sol. $$d = v_0\sqrt{\frac{2\times\frac{d}{2}}{qE_0/m}} = v_0\sqrt{\frac{md}{qE_0}}$$

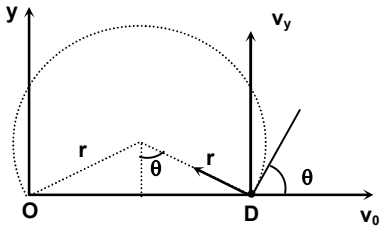
$$E_0 = \frac{mv_0^2}{qd}$$

When the particle strike at D

$$v_y = \sqrt{2\frac{qE_0}{m}\frac{d}{2}} = \sqrt{\frac{qE_0d}{m}}$$

$$d = 2r\sin\theta = 2\frac{mv}{qB}\sin\theta = \frac{2m}{qB}\sqrt{\frac{qE_0d}{m}}$$

$$\Rightarrow B = 2\sqrt{\frac{mE_0}{qd}} = = 2\sqrt{\frac{m}{qd}\times\frac{mv_0^2}{qd}} = \frac{2mv_0}{qd}$$



9. A, C, D

Sol. Current passing through the cell at any time t

$$I = \frac{\varepsilon}{R}\left(1-e^{-Rt/2}\right)+\frac{\varepsilon}{R}e^{-t/RC}$$

$$= \frac{\varepsilon}{R}\left(1-e^{-t/2\tau}\right)+\frac{\varepsilon}{R}e^{-t/\tau} \text{ (where } \tau = RC)$$

I will minimum when $\frac{dI}{dt} = 0$ and $\frac{dI^2}{dt^2} > 0$

$$\frac{dI}{dt} = \frac{\varepsilon}{2R\tau}e^{-t/2\tau}-\frac{\varepsilon}{R\tau}e^{-t/\tau}$$

$$\frac{d^2I}{dt^2} = \frac{\varepsilon}{4R\tau^2}e^{-t/2\tau}-\frac{\varepsilon}{R\tau^2}e^{-t/\tau}$$

$$\Rightarrow \frac{\varepsilon}{2R\tau}e^{-t/2\tau}-\frac{\varepsilon}{R\tau}e^{-t/\tau} = 0$$

$$\Rightarrow e^{t/2\tau} = 2 \Rightarrow t = 2\tau\ell n2 = 2RC\ell n2$$

$$I = \frac{\varepsilon}{R}\left(1-\frac{1}{2}\right)+\frac{\varepsilon}{R}\times\frac{1}{4} = \frac{3\varepsilon}{4R}$$

Charge on capacitor at this time

$$q = \varepsilon C\left(1-e^{-t/\tau}\right) = \varepsilon C\left(1-\frac{1}{4}\right) = \frac{3C\varepsilon}{4}$$

10. A, B, D

Sol. $z_1 = \sqrt{R^2 + X_C^2} = \sqrt{(100)^2 + \left(\frac{1}{50 \times 2 \times 10^{-4}}\right)^2} = 100\sqrt{2}\ \Omega$

$z_2 = \sqrt{R^2 + X_L^2} = \sqrt{(50)^2 + (50 \times 1)^2} = 50\sqrt{2}\ \Omega$

$\tan\phi_1 = \frac{X_C}{R} = \frac{100}{100} \Rightarrow \phi_1 = \frac{\pi}{4}, I_1 = \frac{200}{100\sqrt{2}} = \sqrt{2}\ A$

$\tan\phi_2 = -\frac{X_L}{R} = \frac{-50}{50} \Rightarrow \phi_2 = -\frac{\pi}{4}, I_2 = \frac{200}{50\sqrt{2}} = 2\sqrt{2}\ A$

rms current through ac mains = $\sqrt{I_1^2 + I_2^2} = \sqrt{10}A$

rms potential difference across the inductor = $I_2X_L = 2\sqrt{2} \times 50 = 100\sqrt{2}\ V$

phase difference between the currents passing through the capacitor and inductor = $\frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}$

11. A, C, D

Sol. If the angular position of the point on the screen where phase difference is zero equals θ, then

$\frac{5}{3}d\sin 30° - \frac{25}{18}d\sin\theta = 0$

$\Rightarrow \sin\theta = \frac{3}{5} \Rightarrow \tan\theta = \frac{3}{4}$

$\Rightarrow \frac{y}{1} = \frac{3}{4} \Rightarrow y = 0.75\ m$

If phase difference at point O

$\phi = \frac{2\pi}{\lambda}(\mu d\sin 30°) = \frac{2\pi}{3 \times 10^{-4}} \times \frac{5}{3} \times 1.5 \times 10^{-3} \times \frac{1}{2} = \frac{25\pi}{3}$

Intensity at O, $I = 4I_0\cos^2\frac{25\pi}{6} = 3I_0$

Path difference at O

$\Delta x = \frac{\lambda}{2\pi} \times \frac{25\pi}{3} = \frac{25\lambda}{6}$

The path difference at the nearest minima below O

$\Delta x = \frac{9\lambda}{2}$

$\Rightarrow \frac{25}{18}d\sin\theta = \frac{9\lambda}{2} - \frac{25\lambda}{6} = \frac{\lambda}{3}$

$\Rightarrow \sin\theta = \frac{18\lambda}{75d} = \frac{6}{125}$

Since, $\sin\theta << 1$

$\sin\theta \simeq \tan\theta = \frac{y}{D}$

$\Rightarrow \frac{y}{1} = \frac{6}{125}m \Rightarrow y = 48\ m$

12. C, D

Sol. From conservation of angular momentum

$$mv_0\ell = \left(m\ell^2 + 2m\frac{\ell^2}{3}\right)\omega \Rightarrow \omega = \frac{3v_0}{5\ell}$$

From conservation of mechanical energy

$$\frac{1}{2}\times\frac{5}{3}m\ell^2\left(\frac{3v_0}{5\ell}\right)^2 = mg\ell + 2mg\frac{\ell}{2}$$

$$\Rightarrow v_0 = \sqrt{\frac{20g\ell}{3}}$$

At horizontal position

$\tau = I\alpha$

$$mg\ell + 2mg\frac{\ell}{2} = \frac{5m\ell^2}{3}\alpha$$

$$\alpha = \frac{6g}{5\ell}$$

When the rod is at horizontal position horizontal component of hinge force $F_x = 3m\,a_{cm}$

$$3mg - F_y = 3m\frac{4g}{5}$$

$$F_y = \frac{3mg}{5}$$

SECTION – C

13. 00008.00

Sol. Let the new pressure inside the bubble after expansion be P′

$$P' = \frac{P\frac{4}{3}\pi R_1^3}{\frac{4}{3}\pi R_2^3} = P\frac{R_1^3}{R_2^3}$$

$$P - P' = \frac{\sigma^2}{2\varepsilon_0}$$

$$P - P\frac{R_1^3}{R_2^3} = \frac{q^2}{32\pi^2\varepsilon_0 R_2^3} \quad \Rightarrow q = \sqrt{32\pi^2\varepsilon_0 PR_2\left(R_2^3 - R_1^3\right)}$$

$\Rightarrow n = 8$

14. 00003.11

Sol. For image formed by the refraction at water surface

$$\frac{4/3}{v_1} - \frac{1}{-48} = \frac{\frac{4}{3}-1}{\infty}$$

$\Rightarrow v_1 = -64$ cm

For the lens

$u_2 = -(64 + 11) = -75$ cm

focal length of the lens in water f_w can be found by

$$\frac{f_w}{f_{air}} = \frac{(\mu - 1)}{\left(\frac{\mu}{\mu_w} - 1\right)} = \frac{1.5 - 1}{\left(\frac{1.5}{4/3} - 1\right)}$$

$\Rightarrow f_w = 4f_{air} = 60$ cm

For image formed by the lens

$$\frac{1}{v_2} - \frac{1}{-7.5} = \frac{1}{60}$$

$v_2 = 300$ cm

distance of final image for water surface = 300 cm + 11 cm = 3.11 m

15. 00001.25

Sol. Tension in the string at a distance x from the lower end

$$T = \int_0^x (\mu_0 x dx)g = \frac{\mu_0 x^2}{2} g$$

Speed of wave at this point $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{xg}{2}}$

$$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{gx}{2}} \qquad \Rightarrow t = \sqrt{\frac{2}{g}} \int_0^L \frac{dx}{\sqrt{x}} = \sqrt{\frac{8L}{g}} = \sqrt{8 \times \frac{125}{64}} = 1.25 \text{ sec}$$

16. 00001.00

Sol. Let the speed of the cylinder after a long time be v.

The particle moving from left reach the body with a relative speed $v_0 - v$ and rebound with same relative speed. In a short time interval Δt the particle coming from a distance $(v_0 - v)\Delta t$ reach the cylinder and each imparts an impulse proportional to $2(v_0 - v)$ to the body. Thus the force from the left is proportional to $2(v_0 - v)^2$. Similarity particle in elastically colliding with the right at a negative speed $(v + v_0)$ produce a force proportional to $(v_0 + v)^2$. The condition for constant speed of the body is $2(v_0 - v)^2 = (v_0 + v)^2$

$$\Rightarrow v = \frac{\sqrt{2}-1}{\sqrt{2}+1} v_0 = \left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)(3 + 2\sqrt{2}) = 1 \text{m/s}$$

17. 00003.95

Sol. Energy of the emitted photon = $13.6 \times 3^2 \left(\frac{1}{3^2} - \frac{1}{4^2}\right) \text{eV} = 5.95 \text{ eV}$

Since, $r = \frac{\sqrt{2mK}}{qB}$

Maximum kinetic energy of the photoelectrons

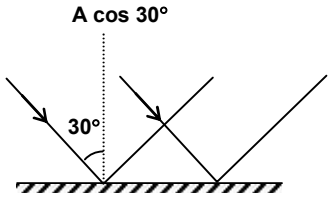
$$K_{max} = \frac{r^2 q^2 B^2}{2m} = \frac{\left(\frac{9.1}{1.6}\right) \times 10^{-4} \times (1.6)^2 \times 10^{-38} \times 4 \times 10^{-8}}{2 \times 9.1 \times 10^{-31}} \text{J} = 2\text{eV}$$

$h\nu = \phi + k_{max}$

$\Rightarrow \phi = h\nu - k_{max} = 5.95 - 2 = 3.95$ eV

18. 00008.00

Sol. I = rate of flow of energy through unit area = $nh\nu$, where n is the no. of photons following through unit area per sec. Rate of flow of momentum (momentum flux) through unit area = $\frac{nh}{\lambda} = \frac{nh\nu}{C} = \frac{I}{C}$.



Consider a cylindrical beam of parallel rays whose section the mirror is A.

Rate of flow of momentum through this beam = $\frac{I}{C}A\cos\theta$

Normal component of the momentum flux = $\left(\frac{IA\cos\theta}{C}\right)\cos\theta$

Momentum flux of the reflected photon = $-0.6\left(\frac{IA\cos^2\theta}{C}\right)$

Force = Rate of change of momentum = $(1.6)\frac{IA\cos^2\theta}{C}$

Pressure = $\frac{F}{A} = \frac{(1.6)I\cos^2\theta}{C} = \frac{1.6\times 2\times 10^3\times\frac{3}{4}}{3\times 10^8} = 8\times 10^{-6}\ N/m^2$

Chemistry

PART – II

SECTION – A

19. B

Sol. $Na_2S + Na_2\left[Fe(CN)_5(NO)\right] \longrightarrow \underset{\text{Purple colour}}{Na_4\left[Fe(CN)_5(NOS)\right]}$

20. C

Sol. $\log\frac{x}{m} = \log k + \frac{1}{n}\log P$

$\log\frac{x}{m} = 0.4 + \log 2$

$\log\frac{x}{m} = 0.4 + 0.3$

$\log\frac{x}{m} = 0.7$

$\frac{x}{m} = 5$

21. B

Sol. $N_3S_{\frac{1}{2}\times 6+\frac{1}{8}\times 6}$

$N_3S_{3+\frac{3}{4}}$

N_4S_5

22. C

Sol. Angular nodes in $4d_{z^2} = 2$

Radial nodes in $4d_{z^2} = 1$

23. A, B, C

Sol. $P_{ext} = 0$ $\quad q = 0$ $\quad \Delta S_{sys} = nC_v\ell n\frac{T_2}{T_1} + nR\ell n\frac{V_2}{V_1}$

$W = -P_{ext}\ \Delta V$ $\quad \Delta U = 0$ $\quad T_1 = T_2$

$W = 0$ $\quad \Delta H = 0$

$T_1 = T_2$ $\quad \Delta S_{surrounding} = 0$ $\quad \Delta S_{sys} = nR\ell n\frac{V_2}{V_1}$

24. A, B, C

Sol. $\underset{\overset{(3)}{3-x}}{X(g)} \longrightarrow \underset{2x}{2Y(g)} + \underset{x}{Z(g)}$

$\frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$

$\frac{1}{2} = \frac{3-x}{2x}\sqrt{\frac{4}{16}}$

$\frac{1}{2} = \frac{3-x}{2x}\times\frac{1}{2}$

X = 2 mole
Y = 2 mole
Z = 1 mole
Total moles = 5
PV = nRT

$P = \frac{5\times 0.08\times 300}{10}$

= 12 atm

$3 - x = 2x$
$x = 1$

$$r = +\frac{dY}{dt} = \frac{(2-0)}{10} = 0.2 \text{ mol } L^{-1} \text{ sec}^{-1}$$

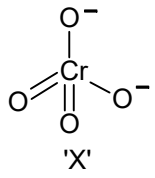
$$K = \frac{1}{10}\ell n\frac{3}{2} = \frac{1}{10}[1.1 - 0.7]$$

$$= \frac{0.4}{10} = 4 \times 10^{-2} / \text{sec}$$

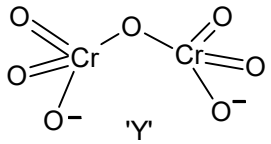
25. A, C

Sol. $FeCr_2O_4 + Na_2CO_3 + O_2 \longrightarrow \underset{'X'}{Na_2CrO_4} + Fe_2O_3 + CO_2$

$Na_2CrO_4 + H^+ \longrightarrow \underset{'Y'}{Na_2Cr_2O_7} + H_2O$



'X'



'Y'

26. A, C

Sol. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$\underset{5-x}{5} \qquad \underset{x}{} \qquad \underset{x}{}$

$m_T = 4 + 5 - x + x + x$

$= 9 + x$

$PV = nRT$

$$n = \frac{4.8 \times 49.2}{0.082 \times 240}$$

$9 + x = 12$

$x = 12 - 9$

$x = 3$

$$\alpha = \frac{3}{5} = 0.6$$

$$K_P = \frac{\frac{3}{12} \times 4.8 \times \frac{3}{12} \times 4.8}{\frac{2}{12} \times 4.8}$$

$$= \frac{3}{12} \times \frac{4.8 \times 3}{2}$$

= 1.8 atm

27. B, C

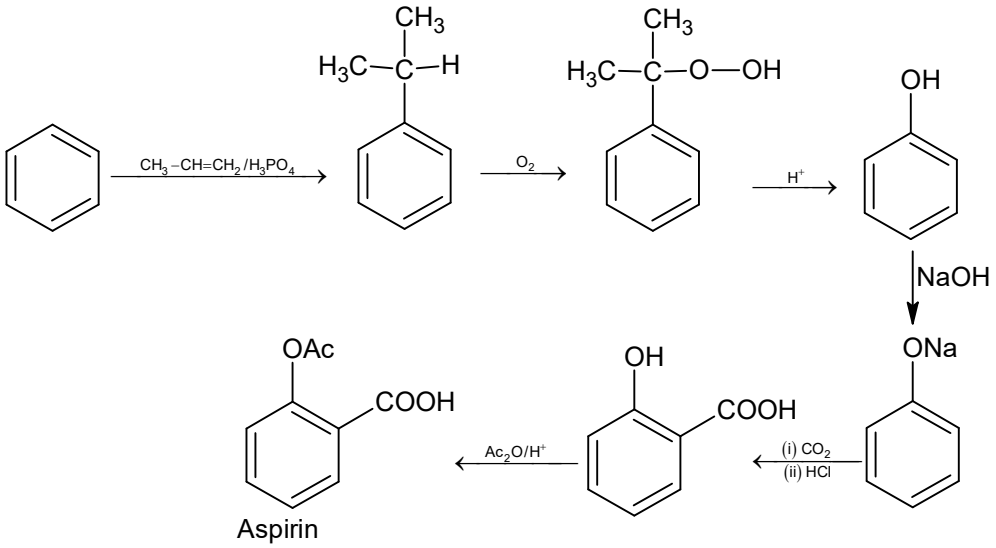
28. C, D

Sol. Sucrose is non-reducing sugar.

α-D-(-) fructofuranose and α-(L)-(+) fructofuranose are enantiomers.

29. A, B, D

Sol.

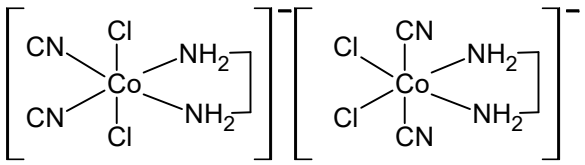


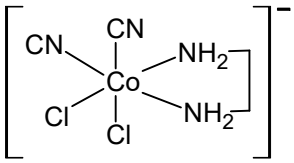
30. A, C

SECTION – C

31. 00004.00

Sol. $\left[Co(CN)_2(en)Cl_2\right]^-$

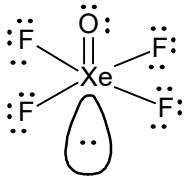




optically active hence mirror image

32. 00014.00

Sol.

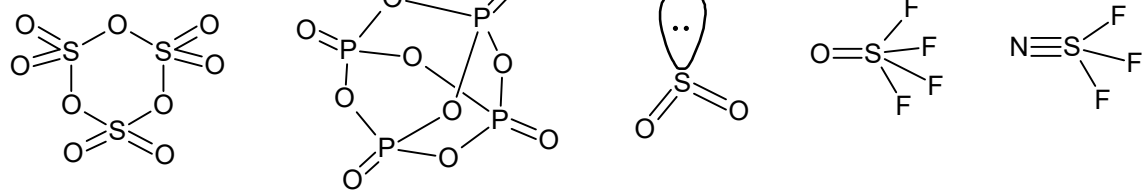


x = 15 x - y = 14

y = 1

33. 00005.00

Sol.



34. 00001.81

Sol. Range 1.81 – 1.82

$$AgCN(s) \rightleftharpoons Ag^{+}(aq) + CN^{-}(aq)$$

$$s - s \qquad s$$

$$y$$

$$Ag^{+} + 2NH_3 \rightleftharpoons \left[Ag^{+}(NH_3)_2\right]^{+}$$

$$s - s \qquad 0.4 - 2s \qquad s$$

$$y \qquad 0.4 - 2s + 2y \qquad s - y$$

$$5 \times 10^{13} = \frac{s}{(0.4 - 2s)^2 \times y}$$

$$5 \times 10^{-3} = s \times y$$

$$25 = \frac{s^2}{(0.4 - 2s)^2}$$

$$5 = \frac{s}{0.4 - 2s}$$

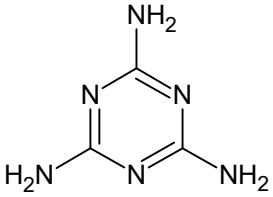
$2 - 10s = s$

$11s = 2$

$$s = \frac{2}{11} = 1.81 \times 10^{-1}$$

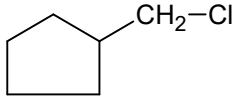
35. 00006.00

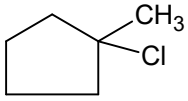
Sol.

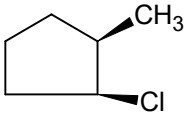


36. 00008.00

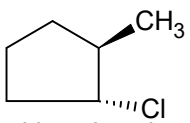
Sol.



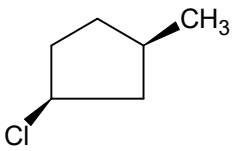


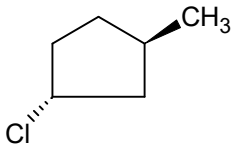


and its mirror image



and its mirror image





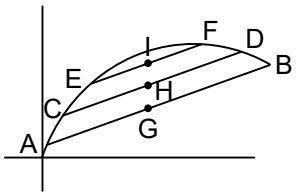
Mathematics

PART – III

SECTION – A

37. B

Sol. Let $y = x^{1/7} \Rightarrow \frac{d^2y}{dx^2} < 0$ for x > 0 and $\frac{dy}{dx} > 0$ for x > 0 $\Rightarrow$ y is increasing and convex graph



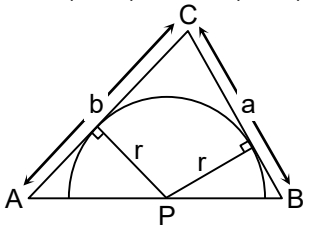
$A\left(2, 2^{1/7}\right)$, $B\left(8, 8^{1/7}\right)$, $E\left(4, 4^{1/7}\right)$, $F\left(6, 6^{1/7}\right)$, $C\left(3, 3^{1/7}\right)$, $D\left(7, 7^{1/7}\right)$

$GI\left(5, \frac{4^{1/7}+6^{1/7}}{2}\right)$, $H\left(5, \frac{3^{1/7}+2^{1/7}}{2}\right)$, $G\left(5, \frac{2^{1/7}+8^{1/7}}{2}\right)$

$\Rightarrow I(5, 2x_3), H(5, 2x_2), G(5, 2x_1) \Rightarrow x_1 < x_2 < x_3$

38. C

Sol. Area (ABC) = Area (APC) + Area (BPC)



$\Delta = \frac{1}{2}(br) + \frac{1}{2}a(r) \Rightarrow r = \frac{2\Delta}{a+b}$

39. D

Sol. Put $x = \sqrt{J}\sqrt{t} \Rightarrow dx = \sqrt{J} \cdot \frac{1}{2\sqrt{t}} dt$

$$\int_0^\infty e^{-x^2} dx = a \Rightarrow \int_0^\infty \frac{e^{-Jt}}{2\sqrt{t}} \sqrt{J}\, dt = a \Rightarrow \int_0^\infty \frac{e^{-Jt}}{\sqrt{t}} dt = \frac{2a}{J} \Rightarrow \int_0^\infty \frac{e^{-Jx}}{\sqrt{x}} dx = \frac{2a}{J}$$

40. C

Sol. Given $A^5 = I = BA^5 = B \Rightarrow (BA)A^4 = B \Rightarrow AB^2A^4 = B$

$\Rightarrow ABAB^2A^3 = B \Rightarrow AAB^2B^2A^3 = B \Rightarrow A^2B^4AA^2 = B \Rightarrow A^2B^3(BA)A^2 = B$

$\Rightarrow A^2B^3AB^2A^2 = B \Rightarrow A^2AB^8A^2 = B \Rightarrow A^3AB^{16}A = B = A^4AB^{32} = B$

$IB^{32} = B \Rightarrow B^{31} = I$

41. A, B

Sol. In all cases both are even. Let x = 2p, y = 2q

$\frac{2}{2p} + \frac{5}{2q} = \frac{1}{6} \Rightarrow 6q + 15p - pq = 0 \Rightarrow (p - 6)(q - 15) = 90$

$\Rightarrow (p - 6)(q - 15) = 2 \times 3^2 \times 5$

(A) When both p and q are even, then (p – 6) is even and (q – 15) is odd so number of solutions
$= 1 \times 3 \times 2 = 6$

(B) p and q both are odd $\Rightarrow$ 6 solutions

(C) p even and q odd $\Rightarrow$ (p – 6)(q – 15) is multiple of 4 which is not true

(D) p odd and q even, then also (p – 6)(q – 15) is multiple of 4 which is not true

42. A, B, C, D

Sol. $\int \frac{9x^9 - 60}{5x^{13} + 3x^{10} - 5x}\,dx = \int \frac{(9x^{-4} - 60x^{-13})}{5 + 3x^{-3} - 5x^{-12}}\,dx$

Put $5 + 3x^{-3} - 5x^{-12} = t \Rightarrow (-9x^{-4} + 60x^{-13})dx = dt = \int \frac{-dt}{t} = \ln|t|$

43. A, B, C, D

Sol. (A) $\sqrt{x} + \sqrt{y} = 1 \Rightarrow x + y + 2\sqrt{xy} = 1 \Rightarrow 4xy = (1 - (x + y))^2$
$\Rightarrow 4xy = 1 + (x + y)^2 - 2(x + y) \Rightarrow (x - y)^2 - 2(x + y) + 1 = 0$ is a parabola

(B) $\frac{1}{x} + \frac{1}{y} = 1 \Rightarrow xy - t - y = 0$ is a hyperbola

(C) $\frac{1}{x} + \frac{1}{y} = \frac{1}{x + y} \Rightarrow t^2 + 3xy + y^2 = 0$ is a pair of lines

(D) $x + y = \frac{1}{x + y} \Rightarrow (x + y)^2 = 0$ pair of lines

44. B, C

Sol. For $n = 41$, $n^2 - n + 41$ is not a prime

(B) $\left(\frac{\ln 2}{1!} + \frac{\ln^2 2}{2!} + \frac{\ln^3 2}{3!} + ...\right) + \frac{\ln 4}{1!} + \frac{\ln^2 4}{2!} + ...$
$= \left(e^{\ln 2} - 1\right) + \left(e^{\ln 4} - 1\right) = 1 + 3 = 4$ = a rational number $\Rightarrow$ B is correct

(C) $T_n = \frac{2n + 1 - 1}{(2n - 1)!} = \frac{1}{(2n)!} - \frac{1}{(2n + 1)!} = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + ... \infty$
$= 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + ... \infty = \frac{1}{e}$, $\ln\left(\frac{1}{e}\right) = -1$ = integer $\Rightarrow$ C is correct

45. A, C

Sol. If we have 4 girls and 100 boys and we want to select 10 students from total of 104 students then the ways will be ${}^4C_1 \times {}^{100}C_9 + {}^4C_2 \times {}^{100}C_8 + {}^4C_3 {}^{100}C_7 + {}^4C_4 {}^{100}C_6 + {}^4C_0 {}^{100}C_{10}$
$= 4\ {}^{100}C_9 + 6\ {}^{100}C_8 + 4\ {}^{100}C_7 + {}^{100}C_6 + {}^{100}C_{10} = {}^{104}C_{10} = {}^{104}C_{94}$
$x + y = 104 + 10 = 114$ or $x + y = 104 + 94 = 198$

46. A, B, D

Sol. $|\bar{a} + \bar{b}| = 1 \Rightarrow 2 = 2\bar{a} \cdot \bar{b} = 1 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 120°$

$(\bar{a} \times \bar{b}) \times \bar{a} = (\bar{a} \cdot \bar{a})\bar{b} - (\bar{a} \cdot \bar{b})\bar{a} = \bar{b} - \frac{1}{2}\bar{a}$

$((\bar{a} \times \bar{b}) \times \bar{a}) \times \bar{a} = \left(\bar{b} - \frac{1}{2}\bar{a}\right) \times \bar{a} = \bar{b} \times \bar{a}$

$(((\bar{a} \times \bar{b}) \times \bar{a}) \times \bar{a}) \times \bar{a} = (\bar{b} \times \bar{a}) \times \bar{a} = (\bar{b} \cdot \bar{a})\bar{a} - (\bar{a} \cdot \bar{a})\bar{b} = \frac{1}{2}\bar{a} - \bar{b}$

$\Rightarrow \bar{\alpha} = \left(\frac{1}{2}\bar{a} - \bar{b}\right) \times \bar{b} = \frac{1}{2}\bar{a} \times \bar{b}$

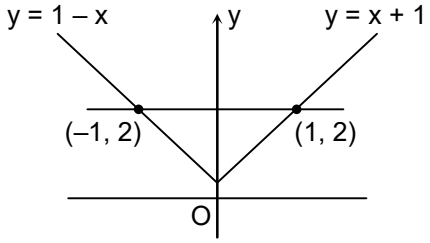
47. B, C

Sol. For $0 \le x \le \pi$, $f(|x|) + |f(x)| = \cos|x| + |\cos x|$, then $g(x) = 0$ for $\frac{\pi}{2} \le x \le \pi$

$\Rightarrow$ Infinitely roots

48. A, C

Sol. From graph it is clear that f(x) is continuous every where but not differentiable at x = 1 and x = –1



SECTION – C

49. 00004.50

Sol. $\frac{4}{\sin x} + \frac{1}{1-\sin x} = 2a$. Let $f(x) = \frac{4}{\sin x} + \frac{1}{1-\sin x}$

$f'(x) = 0 \Rightarrow \sin x = \frac{2}{3}$ least f(x) = $\frac{4}{\frac{2}{3}} + \frac{1}{1-\frac{2}{3}} = 9$

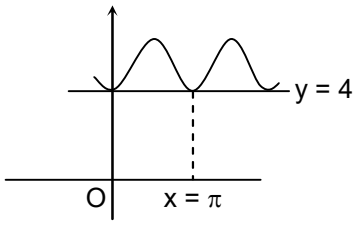
$\sin 2a \ge 9 \Rightarrow a \ge 4.5$

50. 00002.00

Sol. $f'(x) = \begin{cases} \frac{-4}{x^3} + \frac{1}{x^2} & , \quad x < 2 \\ \frac{-1}{x^2} + \frac{4}{x^3} & , \quad x > 2 \end{cases}$ $\Rightarrow$ f(x) is not differentiable at x = 2 also $f'(x) = 0 \Rightarrow x = 4$

51. 00004.00

Sol. Given in the figure



52. 00001.00

Sol. $z_1 + z_2 = -z_3 \Rightarrow |z_1 + z_2| = z_3$

$z_1z_2 + z_2z_3 + z_3z_1 = 0 \Rightarrow |z_3(z_1 + z_2)| = |z_1z_2| \Rightarrow |z_3|^2 = |z_1||z_2|$

53. 00050.00

Sol. $x, x^2 - 24, x + x^2$ are in A.P. $\Rightarrow 2x^2 - 48 = 2x + x^2 \Rightarrow x = 8$ or -6

Possible sequences are $8, 40, 72 \Rightarrow d_1 = 32$

$-6, 12, 30 \Rightarrow d_2 = 18$, sum = 50

54. 00445.00

Sol. Let C_1, C_2, C_3, C_4 be coins P(S) = 1 – coin is not discarded = (1 – P(HH) = $1-\frac{1}{4}=\frac{3}{4}$

Hence, S can take the value 0, 1, 2, 3, 4, P(S = 3 or 4) = P(S = 3) + P(S = 4)

$\Rightarrow \frac{189}{256}=\frac{m}{n} \Rightarrow m + n = 445$